The Path to the Counterexample

Research Progress with AI Assistance

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About This Report

This report accompanies my manuscript *Hierarchical Laminates Cannot Reproduce All Effective Properties of Planar Two-Phase Composites: A Counterexample with a 16-Component Potential* [1]. The research in this manuscript is heavily assisted by OpenAI GPT-6 Astra Ultra. This involves many discussions, computations, and explorations through prompts for a span of two months. Each research finding is recorded, leading to a large number of notes that document the progress and insights gained throughout the project. This report summarizes these notes to make their content accessible to readers.

Once a research question or target theorem is specified, AI systems can now pursue substantial mathematical work autonomously. Examples include Aletheia [2], OpenAI's work on the Navier–Stokes problem [3], its release of more than 700 mathematical manuscripts [4], and its announcement of results on local smoothing, the Exponential Time Hypothesis and quantum field theory [5]. A finished proof can be read and checked while revealing little about the unsuccessful approaches or decisions that led to it. I want to document the full research process and provide transparency. This report provides attribution to the literature that informed the work, an account of unsuccessful approaches and the paths not taken, and the reasons for changes of direction.

The account begins with a review of the connection between two-well relaxation and composite homogenization. It then follows attempts to attain translation bounds, construct periodic gradients for an example with three potentials, and adapt known counterexamples using two quadratic majorants. After examining the limits of duality and other structural reductions, it explains how the constituents and periodic microstructure were chosen together to obtain the counterexample and how that construction was simplified.

Disclaimer. This is an informal account of exploratory work, including abandoned approaches. Its arguments may be incomplete or flawed and are not presented as rigorous mathematical statements or proofs. I prepared this report with OpenAI GPT-6.1 Sol Ultra, using its summaries of a large number of research notes.

1. Two-Well Relaxation and the Composite Problem

The investigation began with Milton's question of whether the quasiconvex and rank-one convex envelopes of the minimum of two strictly convex quadratic functions always

coincide [6, Problem 2.1]. The quadratic functions initially had arbitrary centres and additive constants. The homogeneous quadratic forms used in the manuscript were a later restriction.

Successive binary rank-one splittings of a matrix F produce a probability measure on gradient matrices with barycentre F. Each splitting replaces an atom by two matrices with the same weighted mean and a difference of rank at most one. The resulting measure is called a laminate measure of finite order. More generally, compactly supported laminate measures are characterized by Jensen's inequality for rank-one convex functions; finite support alone does not imply finite order [7, Definition 3.1 and Theorem 4.1].

For the continuous two-well energies considered here, rank-one convexification has the representation

$$\mathrm{R}\,W(F) = \inf_{\nu} \int W(A)\,\mathrm{d}\nu(A),$$

where the infimum is over all laminate measures of finite order with barycentre F, without a bound on the number of splittings [8, Section 2, Lemma 2.2]. A particular splitting measure therefore gives an upper bound for $\mathrm{R}\,W(F)$, whereas a rank-one convex minorant gives a lower bound.

In composite homogenization, a hierarchical laminate is a material formed by successive layering of constituents or previously homogenized composites. Minimizing over compatible gradients in the layers gives its effective energy and enforces continuity of normal flux. For homogeneous wells $q_i(A) = A \cdot L_i A$, the connection with relaxation is

$$\mathrm{Q}\,W(F) = \inf_{\chi} F \cdot B_\chi F, \qquad \mathrm{R}\,W(F) = \inf_{C} F \cdot CF,$$

where χ ranges over periodic phase indicators and C over effective tensors of hierarchical laminates of L_0, L_1. The phase fraction is unrestricted in both infima [6, Section 2(a), equations (2.20)–(2.22)]. Thus optimizing over the layer gradients and then the hierarchical geometry gives the same scalar value as rank-one convexification. Equilibrium enters through this energy minimization; it is not an additional admissibility condition on the splitting measures.

An effective tensor describes the responses at all mean gradients for one microstructure. Equality of the two scalar infima at a prescribed F does not by itself imply equality of the G-closure and lamination closure. The connection between the formulations is used explicitly by Allaire–Lods, who relate the formula for the effective tensor of a sequential laminate to the iteration for rank-one convexification [9, Remark 3.6 and proof of Proposition 3.5]. Milton treats the general relation using additional fields and energy characterizations of tensor sets [6, Section 2(a)].

2. Attaining Translation Bounds by Rank-One Splittings

The translation method suggested a way to prove equality of the envelopes: subtract a quadratic null Lagrangian, optimize the resulting lower bound, and match it by the average energy of a finite-order laminate measure. This approach comes from the variational theory of homogenization of Murat and Tartar and the translation bounds developed by Milton

and Firoozye [10, 11, 12]. It reduced the optimization to finite-dimensional algebra, leaving the construction of rank-one splittings satisfying the equality conditions.

2.1. The Case of Two Potentials

Known results supported this strategy. In his extension from linearized strains to full gradients, Kohn treats two quadratic functions with the same quadratic part [13, Section 7]. Pipkin treats two energies with the same elastic moduli in linearized elasticity [14], while Allaire–Lods obtain equality of the envelopes for an isotropic full-gradient model [9]. These results suggested that rank-one splittings might also attain the quasiconvex envelope for a general minimum of two quadratic functions.

For two potentials, the only quadratic minor is the determinant. The coefficients of admissible determinant translations form an interval. At an interior optimum, stationarity forces the difference between the conditional means assigned to the two wells to have rank at most one, so one rank-one splitting attains the bound. At an endpoint, a vector in the kernel of a translated quadratic form allows another rank-one splitting. The two roots of the compatibility equation then give a finite-order laminate measure with three atoms.

We adapted the endpoint construction from work on planar linearized elasticity by Grabovsky [15, Section 3] and Chenchiah–Bhattacharya [16, Theorem 3.10 and Lemma 3.12]. Their compatible strain jumps are symmetric parts of rank-one matrices; the present problem uses full gradients and ordinary rank-one jumps. The adaptation required treating both endpoints of the translation interval and the case where the translated forms have a common nonzero vector in their kernels. Its success for two potentials encouraged an extension using all the quadratic minors.

2.2. The Difficulty with More Potentials

For more potentials, let $\mathcal{M}(F)$ denote the vector of 2×2 minors of F. These minors are null Lagrangians and provide the quadratic translations [17]. Write the translated quadratic parts as $L_i - C_\tau$, where τ is the vector of translation parameters. An interior optimum still forces the difference between the conditional means assigned to the two wells to have rank at most one.

At a boundary optimum, however, stationarity only gives a relation of the form

$$\mathcal{M}(\Delta F) + \sum_a c_a \mathcal{M}(N_a) = 0, \qquad c_a \geq 0, \tag{1}$$

where ΔF is the difference between these conditional means and each N_a lies in the kernel of one translated quadratic form. The additional terms represent fluctuations of the gradients assigned to each well. These constraints on the minors do not prescribe a sequence of rank-one splittings.

We could attain the bound by rank-one splittings when only one conditional covariance had to be nonzero. An example with three potentials showed that, when both were nonzero, the relation among the minors did not suffice to construct a finite-order laminate measure with the prescribed moments. Perturbing the data did not help: the two constraints expressing singularity were independent and their Lagrange multipliers were positive, so

optima with both constraints active persisted under small perturbations. Continuity from cases with only one singular translated form could not establish the general result.

Failure of rank-one splittings to attain the translation bound did not establish $\mathrm{Q}\,W < \mathrm{R}\,W$: periodic gradients might also fail to attain it. We still needed a periodic gradient with average energy below $\mathrm{R}\,W$ at its mean, either at the translation bound or above it.

2.3. Moment Preservation and Martingales

A parallel approach started with a probability measure on gradient matrices with finite second moments and satisfying the quadratic-minor identities $\mathbb{E}\mathcal{M}(X) = \mathcal{M}(\mathbb{E}X)$. It sought a laminate measure with the same moments relevant to the quadratic energy. The approach used the connection between martingales and convexity along prescribed directions [18], together with constructions of laminate measures using stopped Brownian motion [19]. The decomposition of the covariance followed from the Terpstra–Calderón theorem on nonnegative biquadratic forms with one two-dimensional factor [20, 21].

Let $\iota \in \{0, 1\}$ index the assigned quadratic function. Including ι in the stochastic process allowed us to preserve the masses and conditional means associated with the wells, as well as the total covariance. These moments determine the energy when the quadratic parts agree. Otherwise, the contribution assigned to each well matters. For the fluctuation $Z = X - F$ about the mean, define

$$\Sigma_i = \mathbb{E}\big[\mathbf{1}_{\{\iota=i\}} Z \otimes Z\big]\,, \qquad \Sigma = \Sigma_0 + \Sigma_1.$$

Once the masses and conditional means are fixed, the average quadratic energy is determined by $\operatorname{tr}(L_0\Sigma_0) + \operatorname{tr}(L_1\Sigma_1)$. Preserving Σ alone does not in general preserve this quantity when the quadratic parts differ.

In the simplest Brownian construction, equal probabilities for the wells forced $\Sigma_0 = \Sigma_1$, although the prescribed data need not have this symmetry. A modification aimed to preserve both matrices separately, but the example with three potentials ruled out such a general replacement theorem. Approximating the law of the stopped process, which need not have bounded support, by finite-order laminate measures also remained unresolved. Jensen inequalities for rank-one convex functions of quadratic growth did not by themselves provide that approximation.

Equality of the envelopes did not require preserving all these moments. We therefore tried to preserve the masses and conditional means assigned to the wells while decreasing $\operatorname{tr}(L_0\Sigma_0) + \operatorname{tr}(L_1\Sigma_1)$. One shortcut used an identity expressing energy through a conditional mean, but it applied only to equilibrium fields in the composite problem. Rank-one convexification requires compatible gradient jumps, not equilibrium of the associated flux. Minimizing the energy for a fixed layered geometry would enforce equilibrium but change the conditional means, so the identity did not complete the argument.

Calculations of effective energies at small contrast and for restricted mean gradients produced optimal sequentially laminated microstructures, in agreement with the classical results for ordered constituents [22] [23, Sections 23.3–23.4]. They did not resolve the cases where both translated forms allowed independent fluctuations. We therefore set aside the general construction by rank-one splittings and turned to an explicit counterexample.

3. An Example with Three Potentials

To test whether both conditional covariances could be preserved, we built an example with three potentials using the subspace underlying Šverák's construction [24],

$$E = \left\{ \begin{pmatrix} w_1 & 0 & w_3 \\ 0 & w_2 & w_3 \end{pmatrix} : w \in \mathbb{R}^3 \right\}.$$

Since the rank-one directions in E are its coordinate axes, rank-one convexity within E is equivalent to separate convexity in w. Methods for separately convex hulls [25] therefore allowed us to test whether finite-order laminate measures supported in E could attain the prescribed moments.

3.1. Rank-One Convex Lower Bounds

We chose conditional means F_i and vectors N_i in the kernels of the translated forms so that a probability measure supported on four matrices satisfied the prescribed mean and the identities for the quadratic minors. Its covariance conditional on the well index $\iota = i$ had rank one. Any measure preserving these moments would therefore have to remain on the two affine lines $\mathcal{L}_i = F_i + \mathbb{R}N_i \subset E$.

If a finite-order laminate measure is supported in E, every intermediate matrix, being a conditional average of its final atoms, also lies in E. Every rank-one splitting must therefore follow a coordinate axis. A separately convex function with quadratic growth separated the two lines from the origin. A projection argument extended this separation to sequences of finite-order laminate measures in the full matrix space with bounded second moments, including measures whose atoms only approached the lines.

The same lines defined strictly convex quadratic functions

$$q_i^\varepsilon(F) = \operatorname{dist}(F, \mathcal{L}_i)^2 + \varepsilon |F - F_i|^2, \qquad W^\varepsilon = \min_i q_i^\varepsilon.$$

The separating function gave $\mathrm{R}\, W^\varepsilon(0)/\varepsilon \to \infty$ as $\varepsilon \downarrow 0$. A homogeneous gradient Young measure with mean zero, supported on the lines and generated by gradients with equiintegrable squared norms, would instead give $\mathrm{Q}\, W^\varepsilon(0) = O(\varepsilon)$. It would therefore separate the two envelopes.

The missing step was the existence of this gradient Young measure. The characterization of Kinderlehrer–Pedregal requires Jensen inequalities for quasiconvex functions of quadratic growth, not just the minor identities [26]. Projecting onto two components and applying Müller's theorem on diagonal matrices excluded the original four-atom measure [27], but left open the possibility of other measures on the lines. Thus the example ruled out the proposed replacement by a laminate measure with the same moments without separating the envelopes.

3.2. The Unresolved Construction of Periodic Gradients

For a gradient taking values in E, its three coordinates satisfy

$$w(x, y) = \big(a(x), b(y), c(x + y)\big).$$

The three functions depend on three directions in two spatial variables. The compensated-compactness theorem of Lee, P. F. X. Müller and S. Müller treats a different constraint, in which each component depends on its own independent coordinate [28, Theorem 1.5]. It did not settle whether gradients could approach the union of the two lines in this example.

We also considered whether concentration could help gradients approach the lines. The gradient decomposition lemma of Fonseca, Müller and Pedregal separates an equiintegrable part from a concentrating remainder [29]. Combined with the translation bound, it showed that concentration was unnecessary for attaining the bound. A direct estimate gave a further restriction: the span of N_0, N_1 contains no rank-one direction, so the gradient can be reconstructed from its orthogonal projection onto the complement of that span. Standard Fourier multiplier estimates supplied the required integrability bounds [30]. Concentration at infinity, as considered in [31, 19], could not circumvent this estimate.

Reconstructing the gradient reduced the remaining problem to assigning points in the periodic cell to the two wells, but the energy still contained a trilinear expression in the Fourier coefficients of the corresponding indicator function. It could not be reduced to the positivity of a single Fourier multiplier. Assignments by parallel stripes or by nested stripes at completely separated length scales did not give the required gradient Young measure. In the latter case, the limiting construction was again governed by a martingale with rank-one increments.

A Gram matrix estimate also excluded fields of the form $w = (a(x), b(y), c(x+y))$ from attaining the bound. An extension covered approximate solutions for which, after rescaling, the Fourier coefficients outside a fixed range of frequencies contributed negligibly to the squared L^2 norm. It did not control sequences whose squared L^2 norm was distributed across a widening range of scales.

Several related methods did not resolve this difficulty. The operator describing the three directions does not have constant rank, so the general stability theory of Fonseca–Müller did not apply [32]. Small errors in the differential constraint need not imply strong approximation by an exact solution, as related examples in Kristensen–Raiţă make clear [33, Section 7.1]. Heat-flow arguments motivated by Bruno–Pasqualotto's treatment of the Dacorogna–Marcellini energy did not establish the needed inequality for the separating function [34]. Resonant perturbations and comparisons between martingales and Fourier multipliers suggested further tests [35, 36], but did not produce gradients approaching the prescribed lines. We had ruled out several classes of sequences without constructing a suitable sequence or proving that none could exist.

3.3. A Direct Comparison of the Envelopes

The search could be broadened by allowing periodic gradients to remain at positive distance from the lines. For

$$g(F) = \min_{i} \operatorname{dist}(F, \mathcal{L}_i)^2,$$

a periodic gradient with mean zero and average energy E_{per} would suffice if

$$E_{\mathrm{per}} < c \le \mathrm{R}\, g(0)$$

for some lower bound c. Adding a sufficiently small positive quadratic term to each function would preserve this strict inequality while making both functions strictly convex.

The numerical comparisons discouraged further work on this example. The lower bound for $\mathrm{R}\, g(0)$ remained far below the average energies of the tested periodic gradients. A periodic gradient computed by finite elements had average energy about 0.00676, whereas a finite-order laminate measure with four atoms gave an average of g of about 0.00403, and hence a smaller upper bound on $\mathrm{R}\, g(0)$. Other periodic gradients were not excluded, but finer resolution neither improved the lower bound nor suggested how to obtain a strict inequality.

The proposed rank-one splittings could not attain the translation bound, and no periodic gradient with average energy below $\mathrm{R}\, g(0)$ had been found. This motivated a search in which the periodic field and rank-one convex lower bound could be designed together.

4. Two Quadratic Majorants of Known Nonquasiconvex Functions

Another approach sought two convex quadratic majorants of a known rank-one convex function that is not quasiconvex. The known function would then be a minorant of their minimum. If the average of that minimum along a periodic gradient were below the minorant's value at the mean, the two envelopes would differ. Replacing the known function by this minimum, however, could raise the average energy enough to lose the inequality.

4.1. Why the Known Periodic Gradients Did Not Work

The first candidate was the distribution of a periodic gradient in Milton's sawtooth adaptation of the Šverák and James–Kohn constructions [23, Section 31.8] [24]. This measure has seven atoms. We assigned its atoms to the two quadratic functions and sought a rank-one convex minorant whose value at the mean exceeded the average quadratic energy.

For every assignment of the atoms to the two quadratic functions, we found a finite-order laminate measure preserving each assigned mass and conditional mean while decreasing the conditional second-moment matrices in the positive-semidefinite order. One assignment required such a measure with four atoms to replace the conditional measure on a tetrahedron. Since the average of a quadratic function depends only on first and second moments, neither assigned quadratic energy increased. The resulting upper bound on the rank-one convex envelope was therefore no greater than the average energy of the prescribed periodic gradient, whichever quadratic functions were chosen.

We also tried the periodic gradient formed from three cosines in the work of Agazzi, Bruno, Guerra and Pasqualotto, which adapts Grabovsky's area integrand [35, Lemma 2.1]. Averaging over its four-element symmetry orbits showed that the average of the minimum of any two global quadratic majorants was at least the value of the area integrand at the mean. Adding auxiliary potentials did not help when the rank-one convex minorant depended only on the original potentials: minimizing over the added variables returned two quadratic majorants of the original function. An extension using more potentials would have needed a minorant that also depended on those potentials.

These arguments ruled out the prescribed periodic gradients in the known examples. Their geometry and methods for constructing rank-one convex functions remained relevant, but a counterexample with two quadratic functions would require a different periodic gradient or a different rank-one convex minorant.

4.2. Allowing More General Quadratic Majorants

Grabovsky's representation of quaternionic area as an infimum of positive quadratic energies suggested a further search for two quadratic majorants [37, Section 4, equation (4.9) and Lemma 4.6]. We first restricted the majorants to product form, with positive-definite quaternionic factors acting on the left and right of the matrix. The computed average energy approached, but did not fall below, the area at the mean gradient. Interpreting these homogeneous forms as constituent energies allowed us to apply planar duality: for any pair in this family and every periodic arrangement, the effective energy was at least the area at the mean.

We next constructed quadratic majorants outside that family that attained equality with the area integrand at nonzero matrices. Reducing such a majorant by a constant factor would violate the upper bound at a contact matrix, and the preceding duality argument for product forms no longer applied. Nevertheless, simultaneous optimization over the quadratic functions, periodic gradients, assignments to the two wells and the mean gradient did not produce an average energy below the area at the mean. The searches often returned a constant gradient selecting only one quadratic function.

To remove the restriction to selected families, we used the classical S-lemma in the form of Petersen's lemma [38]. Applying it on quaternionic lines gave a criterion for a homogeneous quadratic function to majorize the area integrand on the whole space. A separate extension argument showed that a strict inequality obtained on a subspace could be preserved when passing to the full matrix space, by adding a small positive quadratic term on the subspace and a sufficiently large term in the transverse directions.

Even after allowing general majorants and extending from subspaces, the search found no periodic gradient with average energy below the minorant at its mean. This did not rule out every pair of majorants, but made further optimization within these examples less attractive.

4.3. What the Numerical Comparisons Changed

Numerical searches also showed the limits of testing only a few rank-one splittings. In one calculation, a fixed checkerboard assignment of the two quadratic functions had lower computed energy than the single splittings tested initially. A finite-order laminate measure with three atoms gave a still lower average. A complementary variational bound then showed that even the minimum over all periodic potentials for that checkerboard assignment exceeded the average for this measure. This excluded the fixed assignment, while leaving open other assignments allowed by the two-well functional.

Computing the rank-one convex envelope on bounded grids produced similar difficulties. An apparent strict inequality survived grid refinement but disappeared when the computational domain was enlarged. This behavior is consistent with the dependence of the

envelope on values outside a local neighborhood, emphasized in numerical approximation theory [8, 39]. In another test, convex combinations of the finite-order laminate measures found by the search, all with the same mean, matched the prescribed moments more closely than any of those measures alone.

We therefore turned to the composite problem: first establish an identity or inequality for the effective tensor of every hierarchical laminate of the chosen constituents, then find a periodic microstructure violating it. This would cover all layer normals, proportions and numbers of levels before converting the separation of effective tensors into a two-well energy.

5. Structural Reductions and Their Limits

We also studied the related composite problem, seeking changes to the constituent tensors that would allow us to apply known bounds attained by hierarchical laminates. These methods concerned effective responses, rather than directly computing the envelopes of arbitrary two-well integrands.

Planar duality was a natural candidate because it exchanges gradient and complementary energies and underlies many exact relations [40, 41] [23, Chapter 3]. When the difference between the constituent tensors was indefinite, common changes of potential variables, planar duality and admissible translations by a quadratic null Lagrangian did not make the pair ordered. Dualizing only some fields introduced both gradient and flux variables and, for general coupling, no longer gave a symmetric positive-definite constitutive law [23, Section 11.5]. These transformations therefore did not extend the known bounds for ordered constituents to the original pair.

The quaternionic family in Grabovsky's example is preserved by lamination but not by homogenization [42, 37]. Milton had already considered selecting two members as constituents and stated that the determinant relation would remain valid [6, Remark 2.4]. A direct duality argument confirmed this and covered any three selected members: their imaginary quaternion coordinates lie in an affine plane, and translation by a common quadratic null Lagrangian followed by planar duality preserves the relation under homogenization. A separate algebraic construction with two generators was blocked, in that representation, by Cohn's reversibility theorem and the completeness criterion for Jordan algebras [43] [42, Section 2.2]. These arguments concerned only the tested families and did not establish Milton's broader assertion for every family preserved by lamination.

Operator representations offered another possible reduction. Methods based on successively removing layers give representations of effective conductivity functions for certain scalar and polycrystalline problems [23, Section 18.5] [44]. Rational matrix-valued Stieltjes functions also admit continued-fraction reductions [45]. We investigated whether these representations could be used to construct hierarchical laminates for the coupled system.

The difficulty was realizing all fields in the same microstructure. Each layer has one normal direction shared by every potential, and every branch of the hierarchy must end at one of the same two constituents. The operator representations need not preserve these constraints. For example, separate representations could assign different effective tensors to

two identical uncoupled fields, although no common microstructure could give those tensors. The representations alone therefore did not resolve the geometric realization questions discussed in the literature [46, 47]. Rewriting equilibrium fluxes in terms of stream potentials led to related subspace problems, but we found no theorem constructing the necessary laminates.

6. Designing the Constituents

The next approach was to choose two constituents so that every hierarchical laminate formed from them would satisfy the same energy equality. To obtain a counterexample, it would then suffice to find a periodic microstructure whose effective tensor violated that equality.

6.1. Why Families Preserved by Lamination Were Useful

This approach follows Francfort–Milton's study of sets of tensors preserved by lamination and the theory of exact relations developed by Grabovsky, Milton and Sage [48, 49]. Grabovsky used a family preserved by lamination but not by homogenization to construct a rank-one convex function that is not quasiconvex [42, 37]. The task was to obtain such a failure of preservation under homogenization from just two constituents.

The formula for layering uses one normal direction at each step, whereas the expansion for a periodic medium contains products of matrices associated with several Fourier directions. The criteria for exact relations reflect this distinction: preservation under lamination imposes weaker algebraic conditions than preservation under homogenization, which also involves products associated with different directions [49, Theorems 3.5–3.6 and 3.9].

Although the phase contrast is fixed, the matrices $T(k)$ vary with direction. The broader preservation statement in Milton's Remark 2.4 would rule out the counterexample [6], but a fixed contrast does not imply the conditions involving different directions in the theorem on exact relations. We therefore tested those conditions directly.

6.2. Choosing the Matrix Space

We chose the representation of a spin factor from Grabovsky's classification of the relevant Jordan algebras [42, Theorems 4.3 and 6.7]. The space $\mathcal{S}$ is closed under the inversions and averages in the lamination formula. However, the matrix

$$\Omega = \Gamma_1\Gamma_2\Gamma_3\Gamma_4 \notin \mathcal{S}$$

is symmetric. Products involving different directions can generate a term proportional to Ω, although the reduced matrix H representing the effective tensor of each hierarchical laminate remains in $\mathcal{S}$.

We chose P, D so that $T(k) \in \mathcal{S}$ for every direction. The distinct vertical coefficients in D make the coefficients of $T(k)$ vary with direction, while the factorization defining P keeps the matrices in $\mathcal{S}$. The transformation used in the lamination formula comes from the theory of exact relations [49, Theorem 3.1]. With these choices, the compact family $\mathcal{K}$ is preserved under layering for every normal and choice of proportions.

We chose the prescribed mean gradients ξ, η so that every tensor in $\mathcal{K}$ gives equal energies at them. Thus the effective tensor C of every hierarchical laminate satisfies $\ell(C) = 0$. A periodic medium with effective tensor B satisfying $\ell(B) > 0$ would therefore place B outside the lamination closure. Expressing the separation through two energies would later reduce the number of potentials needed for the two-well application.

Products of at most three matrices $T(k)$ give no difference between the selected energies, whereas four can produce a term proportional to Ω. We therefore sought a positive fifth-order coefficient in the weak-contrast expansion. This reduced the problem to choosing a trigonometric polynomial for the local phase fraction and computing that coefficient.

7. Completing and Simplifying the Counterexample

We first sought a violation of the energy equality in a smooth coefficient field whose values were local effective tensors of the two constituents. This allowed the energy difference to be calculated before replacing the local mixtures by a two-phase microstructure at a smaller scale.

7.1. Choosing the Spatial Variation

For these constituents, horizontal layers with fraction t of phase 1 have effective tensor $(1-t)L_0+tL_1$. Varying the local phase fraction t in space therefore gave a smooth coefficient field whose values could later be realized by layers of the same two constituents.

We studied the effective tensor using the standard weak-contrast expansion [23, Sections 14.1 and 14.6], whose ordered products of matrices $T(k)$ also appear in the criteria for exact relations [49]. The task was to choose t so that the coefficient of ρ^5 in $\ell(B)$ was positive.

The first construction used five cosine terms. The simpler profile in the manuscript uses three cosine terms with frequencies

$$a = (1, 0), \qquad b = (0, 1), \qquad c = (1, 3), \qquad a + 3b - c = 0.$$

The frequencies $a, b, b, b, -c$ sum to zero, allowing the product of their five Fourier coefficients to contribute to the average at fifth order. The term contains four matrices $T(k)$. Since $T(b) = 0$, most orderings contribute zero. Four contributions survive and agree, with a positive sign given by the Cauchy determinant formula.

The use of a few interacting Fourier modes has precedents in Šverák's construction and the resonant perturbations of Agazzi and coauthors [24, 35]. Convergence of the series and a bound on the remainder ensured that the positive fifth-order term gave $\ell(B) > 0$ for sufficiently small positive contrast. The sign no longer depended on numerically resolving a small energy difference.

7.2. Retaining the Inequality with Two Phases

The smooth coefficient field took continuously many values. To replace it by a two-phase microstructure, we used the principle of locally periodic homogenization [50, 51], with horizontal stripes whose widths followed the local phase fraction.

Within each thin horizontal strip, we replaced $t(x, y)$ by its value at one vertical coordinate. Integrating the difference between the resulting phase indicator and this sampled fraction in the vertical direction gave a function that vanished at both strip edges. Its derivatives supplied a correction to the flux that preserved equilibrium while leaving a small residual in the constitutive relation. Coercivity then bounded the error in the mean flux at ξ and η.

We first chose the contrast so that $\ell(B) > 0$ for the smooth medium, then made the stripes fine enough that the error in this difference was smaller than $\ell(B)$. This gave a periodic two-phase microstructure with finitely many stripes and an effective tensor outside the lamination closure. Only convergence of the mean fluxes at ξ, η was needed, rather than convergence of the full tensor.

7.3. Reducing the Number of Potentials

The first complete construction used forty scalar potentials. A general conversion from separation of effective tensors to a two-well energy used eighty copies of this system for gradient energies and eighty for complementary energies, giving 6400 potentials. It followed the established use of these two kinds of energy to distinguish effective tensors [52, Remark 3.7] [23, Section 30.3], but represented many more energies than the separation required.

Factoring each matrix $I_8 + \Gamma_j$ through $\mathbb{R}^4$ reduced the constitutive system to sixteen scalar potentials while retaining the equality for hierarchical laminates and its violation by the periodic composite. Choosing three cosine terms for the local phase fraction reduced the fifth-order calculation to the four contributions above.

The energy equality gave a larger reduction in the two-well application. The constituents are ordered, so the pointwise minimum of their gradient energies is one quadratic form, but their complementary energies have the opposite order. Since the separation involved only the energies at ξ, η, one gradient energy and one complementary energy sufficed. Neither resulting quadratic function dominates the other; the two-well terminology refers to these two branches, which share their minimum at the origin. Minimizing the sum of gradient and complementary energies over $C \in \mathcal{K}$ gave a function below both forms. Preservation of $\mathcal{K}$ under lamination and planar duality proved it rank-one convex, following [52, Remark 3.7] and [37, Theorem 4.1]. Both groups of potentials used the same phase indicator, so the energies came from one microstructure.

The resulting two-well energy uses thirty-two potentials. Prescribing the overall phase fraction or adding uncoupled potentials did not change the argument. The construction first separated an attainable effective tensor from the lamination closure, then converted that separation into a gap between the quasiconvex and rank-one convex envelopes.

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